

CLIMATE DATASETS · free for noncommercial use

THEME	DATASET ID	RES	KEY BAND
● Rainfall	UCSB-CHG/CHIRPS/DAILY	5 km	precipitation
● Rainfall (v3)	UCSB-CHC/CHIRPS/V3/DAILY_SAT	5 km	precipitation
● Air temp	ECMWF/ERA5_LAND/DAILY_AGGR	11 km	temperature_2m
● Land heat	MODIS/061/MOD11A1	1 km	LST_Day_1km
● Sea-surf temp	NOAA/CDR/OISST/V2_1	0.25°	sst, anom
● Elevation	NASA/NASADEM_HGT/001	30 m	elevation
● Surface water	JRC/GSW1_4/GlobalSurfaceWater	30 m	occurrence
● Boundaries	USDOS/LSIB_SIMPLE/2017	vec	country_na

UNIT GOTCHAS · apply before mapping

- ERA5 & MODIS temperatures are in KELVIN -> subtract 273.15 for °C
- MODIS LST: multiply by 0.02 first, then convert from Kelvin
- OISST 'sst' and 'anom': multiply by 0.01 to get °C
- CHIRPS v2 ends after Dec 2026 -> switch to v3 for future work
- Ocean data (OISST): use a buffered offshore area, not the land outline

SIGN UP IN 5 STEPS · one time, free, no credit card

- 1 Go to code.earthengine.google.com/register and sign in with Google
- 2 Create a Cloud project (e.g. 'Earth Engine Samoa') — No organization
- 3 Verify eligibility: 'Earth Engine trainer or trainee' -> role 'Participant'
- 4 Choose the free Community tier; describe work = 'Classroom or education'
- 5 Register -> open the Code Editor; your project shows top-right
Re-verify your noncommercial status once a year.

THE CORE WORKFLOW



LOCALISE FOR YOUR COUNTRY

Set the friendly name in any script:

```
COUNTRY = 'Solomon Islands'
```

The script's selector picks the right area automatically:

- Big islands = real LSIB outline: Fiji, Samoa, Vanuatu, Solomon Is, Papua New Guinea, New Caledonia (note State Dept spelling 'Solomon Is')
- Small/atolls = point + buffer: Tonga, Palau, Tuvalu, Kiribati, Nauru, Niue, Cook Is, Marshall Is, Micronesia (LSIB_SIMPLE drops these)

Confirm live: run 99 diagnostic check.js once after signup.